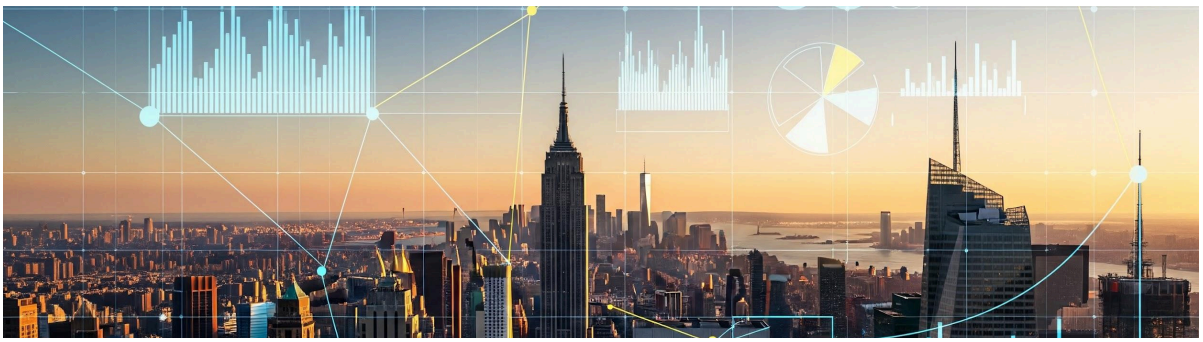

NYC Real Estate Sales Analysis: Capstone Project Report

Project Title: NYC Real Estate Sales Analysis

Sector: Real Estate

Team: Shadow-InsightX

Team Members: Anuj Chhajed, Amrit Kumar Mahto, Nakul Sharma, Ayush Kumar, Parth Tandalwade

Github URL: https://github.com/THE-Amrit-mahto-05/Shadow-InsightX

Tableau Public Dashboard URL:

Institute: Newton School of Technology

Submission Date: 28 Apr 2026

Executive Summary

Problem: The New York City real estate market — one of the world's largest — remains highly opaque for investors, developers, and policymakers. Property valuations vary dramatically across boroughs, building types, and seasons, yet most market participants rely on city-wide averages that mask these differences. This project set out to answer a critical **question**: *What actually drives property prices in NYC, and how can stakeholders make smarter decisions with data?*

Approach: We analysed over 84,000 property transactions from the NYC Department of Finance's Rolling Sales dataset. The raw data was cleaned and enriched using Python (Pandas) — removing non-market transfers, handling

outliers, and engineering new metrics such as Price Per Square Foot and Building Density (Floor Area Ratio). We then conducted rigorous statistical testing (ANOVA, T-Tests, Regression) to validate patterns, and built an interactive Tableau dashboard for stakeholder exploration.

Key Insights:

1. **Location is the single biggest price driver.** A square foot of space in Manhattan costs 3× more than the same space in the Bronx ($695 vs. $226) — a gap that is statistically confirmed, not anecdotal.
2. **Residential and commercial properties are fundamentally different markets.** Commercial properties average $2.89M vs. $917K for residential. Mixing them in any analysis produces misleading conclusions.
3. **Seasonal patterns affect transaction volume, not price.** June sees 37% more sales than February, yet median prices remain stable year-round. Sellers don't discount in slow months — they simply wait.
4. **Building age doesn't reduce value.** Over 42% of NYC's housing stock is 80+ years old, yet older buildings in prime locations command the same premiums as new construction.

Key Recommendations:

1. **Adopt borough-specific pricing models.** City-wide averages are misleading. Investment and valuation decisions should use localized benchmarks — a property priced at $400/sqft is undervalued in Manhattan but overvalued in the Bronx.
2. **Time property listings for the April–July window** to maximise exposure during peak buyer activity, while buyers seeking leverage should target the quieter January–February period.

Separate residential and commercial portfolios in all reporting, valuation, and risk assessment to avoid cross-contamination of fundamentally different market dynamics.

Sector & Business Context

Sector Overview and Current Industry Challenges

The U.S. real estate sector contributes over $4.5 trillion annually to the national GDP, and New York City sits at its epicentre — processing tens of thousands of property transactions each year across five boroughs with vastly different market dynamics. Despite this scale, the industry faces a persistent transparency problem. Property valuations are still heavily influenced by subjective appraisals, broker intuition, and inconsistent comparables. Key challenges include:

- **Information asymmetry:** Buyers and sellers operate with unequal access to

pricing data, neighbourhood trends, and property-level metrics — leading to mispricing and missed opportunities.
- **Market fragmentation:** NYC's five boroughs function as semi-independent markets. A city-wide median price of $636K masks the reality that Manhattan's price per square foot ($695) is 3× the Bronx's ($226). Stakeholders who rely on aggregated figures make systematically flawed decisions.
- **Non-market noise in public data:** A significant portion of recorded transactions are $0 transfers (ownership changes, tax arrangements) that distort any analysis performed on raw data without careful cleaning.

Who Is the Decision-Maker This Project Serves?

This project is designed for four primary stakeholder groups:

1. **Real estate investors** seeking data-backed location and timing strategies to maximise returns across NYC's fragmented borough markets.
2. **Property developers** evaluating land acquisition opportunities — particularly those deciding between new construction and pre-war building renovation in high-demand boroughs.
3. **Financial institutions** (banks, mortgage lenders) that need reliable, quantified valuation benchmarks for collateral assessment and portfolio risk management.
4. **City planners and tax authorities** responsible for ensuring equitable property tax assessments and informed zoning policy across neighbourhoods with 3×–5× price variation.

Why Was This Specific Problem Chosen?

The NYC Rolling Sales dataset is one of the richest publicly available real estate datasets in the world — yet it is rarely analysed beyond surface-level reporting. Most existing analyses treat NYC as a single market, ignore the impact of building class and density on pricing, and fail to separate residential from commercial transactions. This creates a gap between available data and actionable intelligence.

We chose this problem because:

- The data is publicly available but underutilised — a classic opportunity for analytics to create value.
- The market's complexity (5 boroughs, 30+ building classes, extreme price variance) makes it an ideal candidate for statistical segmentation and hypothesis testing.
- The findings are directly actionable — every insight maps to a concrete investment, policy, or operational decision.

What Is the Business Value of Solving It?

Solving this problem delivers measurable value across the stakeholder chain:

- **For investors:** Borough-specific PPSF benchmarks ($226–$695/sqft) replace guesswork with precision. The seasonal analysis (37% volume surge in June, stable prices year-round) provides a clear listing-timing advantage.
- **For developers:** The finding that building age has near-zero correlation with price (r = +0.017) — while building density (FAR) shows moderate positive

correlation (r = +0.51) — redirects capital toward high-FAR renovation opportunities instead of speculative new construction.
- **For lenders:** The regression model's RMSE of $1.12M and R^2 ceiling of 0.37 quantify the risk of relying on linear valuation models, directly informing the transition to more robust, segmented approaches.
- **For policymakers:** Statistically validated price disparities (ANOVA F = 2,607 across boroughs) provide the evidence base needed for equitable tax assessment reform.

In short, this project converts a raw public dataset into a strategic decision-support tool — replacing assumptions with evidence at every level of the value chain.

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1. Problem Statement

The New York City real estate market is one of the world's most dynamic and least transparent. Potential investors, developers, and policymakers lack a clear, data-driven understanding of the key factors influencing property valuation and sales volume. The core business problem is the high level of uncertainty in determining *fair market value* and identifying *high-potential investment areas*.

Objectives:

1. **Identify Key Value Drivers:** Determine which property attributes (e.g., location, size, building class, density) have the most significant impact on Sale Price.
2. **Analyse Market Segmentation:** Segment the market by borough, property type, and price bracket to understand unique sales trends and pricing dynamics.
3. **Develop Predictive Insights:** Build baseline regression models and provide data-backed recommendations to guide investment strategies, development decisions, and tax assessment policies.

Scope:

- **In Scope:** Cleaning and analysis of the NYC Rolling Sales dataset (rolling 12-month period: Sep 2016 – Aug 2017); EDA, hypothesis testing (ANOVA, T-Test, Chi-Square), and regression modelling in Python; interactive Tableau dashboard; business recommendations.
- **Out of Scope:** Real-time price prediction, external macroeconomic data integration (interest rates, inflation), geospatial modelling (subway proximity, school districts), and non-linear/ensemble modelling (Random Forest, XGBoost).

Success Criteria:

1. Statistically validated relationships between property attributes and sale price

($p < 0.05$).
2. A baseline regression model with quantified predictive accuracy (R^2 and RMSE).
3. At least 5 actionable, data-backed business recommendations linked to specific analytical findings.
4. An interactive Tableau dashboard enabling stakeholder-driven exploration by borough, property type, and time period.

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2. Dataset Description

The analysis is based on the **NYC Rolling Sales** dataset, a publicly available record of property transactions.

Source of Dataset:
https://www.kaggle.com/datasets/new-york-city/nyc-property-sales

Metric	Value
Initial Number of Rows	~84,000
Initial Number of Columns	21
Time Period Covered	Rolling 12 months (e.g., 🗓 Date to 🗓 Date)

Key Features:

- **BOROUGH:** The administrative division of NYC (1-5).
- **BUILDING CLASS CATEGORY:** High-level grouping of property types (e.g., 01 ONE FAMILY DWELLINGS).
- **RESIDENTIAL UNITS / COMMERCIAL UNITS:** Number of units in the building.
- **GROSS SQUARE FEET:** Total interior area of the building.
- **SALE PRICE:** The transaction price of the property (the key dependent variable).
- **SALE DATE:** Date of the transaction.

Data Limitations:

1. **Non-Market Transactions:** The dataset contains numerous $0 sale prices representing ownership transfers, tax arrangements, and non-arm's-length transactions — not fair market value.
2. **Missing Square Footage:** 39,029 rows (46%) have missing or zero GROSS SQUARE FEET, limiting the usable subset for price-per-sqft analysis.

3. **Inconsistent Labelling:** The NEIGHBORHOOD and BUILDING CLASS fields contained inconsistencies requiring standardisation.
4. **Missing Context:** The dataset lacks external economic indicators (interest rates, inflation, employment), internal property condition data (renovation status, amenities), and geospatial features (subway proximity) — all significant price drivers.

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3. Data Cleaning & ETL Process

All primary cleaning and transformation steps were executed in **Python (Pandas, NumPy)** within a Jupyter Notebook. **Reference:** notebooks/02_cleaning.ipynb (committed to GitHub)

Step 1 — Column Standardisation:

- Renamed all 21 columns from uppercase with spaces to snake_case (e.g., SALE PRICE → sale_price).
- Dropped 5 irrelevant columns: ease_ment (100% blank), address, apartment_number, block, lot.

Step 2 — Data Type Conversions:

- sale_price, land_square_feet, gross_square_feet were stored as strings with dash (" - ") placeholders → converted to numeric (float64), coercing invalid values to NaN.
- sale_date → converted to datetime. Validated range: Sep 1, 2016 – Aug 31, 2017.
- year_built → replaced 6,970 zero values with NaN (invalid construction years).

Step 3 — Handling Missing Values:

Column	Missing Count	Treatment
sale_price	14,561 rows	Dropped (missing = non-transaction)
gross_square_feet	39,029 rows	Zeros → NaN; rows retained but excluded from PPSF calculations
land_square_feet	36,578 rows	Zeros → NaN; rows retained but excluded from FAR calculations

year_built	6,970 rows	Zeros → NaN; rows retained for non-age-dependent analysis
tax_class_at_present	584 rows	Blank strings → NaN; rows retained

Step 4 — Filtering Invalid Sales:

- Removed 14,561 rows with missing sale_price (dash placeholders).
- Removed 10,228 rows with $0 sale_price (non-market transfers).
- Removed 1,002 rows with sale_price ≤ $100 (non-arm's-length transfers).

Step 5 — Outlier Treatment:

- **Sale Price:** Filtered rows outside 1st and 99th percentiles ($25,000 – $14,000,000) → removed 1,156 extreme outliers.
- **Gross Square Feet:** Capped at 99th percentile (27,598 sqft) to limit influence of mega-properties.

Step 6 — Feature Engineering (10 new columns):

Feature	Formula / Logic	Purpose
price_per_sqft	sale_price / gross_square_feet	Normalised property valuation metric
price_per_unit	sale_price / total_units	Per-unit investment cost for multi-family
building_age	sale_year - year_built	Property age at time of sale
floor_area_ratio	gross_square_feet / land_square_feet	Building density indicator
is_residential	1 if residential_units > 0, else 0	Residential/commercial flag
has_units	1 if total_units > 0, else 0	Vacant lot identifier
size_category	Binned gross_sqft: Small/Medium/Mid-Large/ Large/Very Large	Property size segmentation

price_bracket	Binned sale_price: <250K / 250K-500K / 500K-1M / 1M-2.5M / 2.5M-5M / 5M+	Price tier segmentation
building_class_code	Extracted numeric code from category	Clean classification code
building_class_description	Extracted text description from category	Readable property type
borough_name	Mapped 1→Manhattan, 2→Bronx, 3→Brooklyn, 4→Queens, 5→Staten Island	Human-readable borough
sale_year/month/quarter/day_of_week	Extracted from sale_date	Temporal analysis features

Assumptions Made During Cleaning:

1. All $0 and ≤$100 sales are non-market transactions and were excluded entirely (not imputed).
2. Zero values in gross_square_feet and land_square_feet represent missing data, not actual zero-area properties.
3. Negative building_age values (year_built > sale_year) are data entry errors and were set to NaN.
4. The 1st–99th percentile range ($25K–$14M) adequately captures the legitimate NYC market while removing data errors.

Final Cleaned Dataset: 84,548 → **57,601 rows** | 21 → **31 columns**

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4. KPI Framework

Key Performance Indicators (KPIs) were defined to focus the analysis on metrics directly relevant to investment and market health.

Valuation Metrics:

KPI	Definition	Importance
Median Sale Price	Middle value of all property transactions ($636K).	More reliable than mean for skewed markets — represents

KPI	Definition	Importance
		what a "typical" buyer actually pays.
Price Per Square Foot (PPSF)	`sale_price / gross_square_feet` (Median: $350).	Normalizes property value by size, enabling fair comparison across different property types and boroughs.
Price Per Unit	`sale_price / total_units` (Median: $491K).	Measures per-unit investment cost — critical for multi-family investors evaluating rental income potential.

Market Activity Metrics:

KPI	Definition	Importance
Monthly Transaction Volume	Count of sales per month (Range: 4,276 – 5,841).	Tracks market liquidity and demand cycles — peak months signal competitive buying windows.
Borough Market Share	% of total transactions per borough (Queens: 31%, Brooklyn: 26%, Manhattan: 24%).	Identifies where market activity is concentrated vs. underserved — guides resource allocation for agents and developers.
Price Bracket Distribution	% of sales in each price tier (62% fall in $250K–$1M).	Defines the addressable market size at each price level — essential for targeting and segmentation.

Property Characteristic Metrics:

KPI	Definition	Importance
Building Age	sale_year - year_built (Median: 68 years).	Flags renovation needs and historical value premiums —

KPI	Definition	Importance
		42.5% of stock is 80+ years old.
Floor Area Ratio (FAR)	`gross_square_feet / land_square_feet` (Manhattan median: 2.86).	Measures building density — higher FAR correlates with higher prices (r = +0.51) and indicates development potential.

Model Performance Metrics:

KPI	Definition	Importance
R-squared (R^2)	Proportion of price variance explained by the model (Best: 0.37).	Measures predictive accuracy — current ceiling of 37% confirms need for advanced modeling approaches.
RMSE	Root Mean Squared Error of predictions ($1.12M).	Quantifies average prediction error in dollar terms — directly translates to business risk in valuation estimates.

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5. Exploratory Data Analysis (EDA)

EDA revealed fundamental patterns in the NYC real estate market, setting the stage for deeper statistical analysis, leading to the following key insights:

1. **The "average" NYC property doesn't exist.** The mean sale price ($1.07M) is 67% higher than the median ($636K), revealing a market heavily distorted by luxury transactions in the upper tail.
2. **Manhattan charges 3× the Bronx for the same square footage.** Median price per sqft is $695 in Manhattan vs $226 in the Bronx — location alone triples the cost of space.
3. **62% of all transactions happen in the $250K–$1M band.** The bulk of the market is standard residential. The luxury segment (>$2.5M) represents only 8% of transactions but disproportionately skews market averages.

4. **Property size scales price predictably, but location caps the ceiling.** A "Very Large" property's median price is $4.98M, yet a "Large" property in the Bronx still sells for less than a "Medium" one in Manhattan.
5. **Summer drives volume, not price.** June sees 37% more transactions than February (5,841 vs 4,276), yet median prices remain stable ($600K–$714K year-round). Sellers hold through slow months rather than discounting.
6. **42.5% of properties are over 80 years old, yet age doesn't hurt value.** Building age has near-zero correlation with price (r = +0.017). Pre-war buildings in prime locations retain — and often command — luxury premiums.
7. **Manhattan's building density (FAR) is 5.6× Staten Island's.** Manhattan's median Floor Area Ratio is 2.86 vs 0.51 for Staten Island, and FAR correlates positively with sale price (r = +0.51) — denser buildings are worth more.
8. **Non-residential properties sell at lower median prices but higher extremes.** Residential median ($694K) exceeds non-residential ($465K), but commercial outliers push the non-residential mean far higher — two fundamentally different markets sharing one dataset.
9. **Thursday is the most common closing day.** 24% of all sales close on Thursdays, with Monday being the slowest. This reflects business-hour legal/financial closing patterns, not buyer preference.
10. **Square footage is the strongest single predictor of price (r = +0.59), but still leaves 41% of variance unexplained.** Size is necessary but not sufficient — any predictive model must incorporate location, building class, and density features to be accurate.

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6. Statistical Analysis

The statistical analysis focused on validating key relationships, testing distributional assumptions, and building baseline predictive models.

Correlation Analysis — NYC Rolling Sales:-

Rank	Feature	Correlation (r)	Strength
1	price_per_unit	+0.603	Moderate-Strong
2	gross_square_feet	+0.594	Moderate-Strong
3	floor_area_ratio	+0.509	Moderate
4	price_per_sqft	+0.436	Moderate
5	residential_units	+0.137	Weak
6	building_age	+0.135	Weak
7	total_units	+0.114	Weak
8	land_square_feet	+0.046	Negligible
9	commercial_units	+0.016	Negligible

price_per_unit and gross_square_feet are nearly tied as the strongest linear predictors, while building age and unit counts show minimal linear impact.

Normality Tests (Shapiro-Wilk):-

Variable	p-value	Normal?
sale_price	1.36e-81	❌ No
price_per_sqft	7.97e-84	❌ No
gross_square_feet	3.15e-84	❌ No
building_age	2.00e-46	❌ No

No variable is normally distributed; all are right-skewed. While ANOVA remains valid due to large sample sizes (Central Limit Theorem), this confirms that log-transforms or

non-parametric methods would improve subsequent modeling.

Hypothesis Tests:-

Test 1 — ANOVA: Does borough affect sale price?

- **F-statistic:** 2,607.62
- **p-value:** ≈ 0 (essentially zero)
- **Verdict:** ✅ **Reject H_0** — Borough creates massive, statistically significant price differences.

Test 2 — T-Test: Residential vs Non-Residential

- **t-statistic:** −20.86
- **p-value:** 1.35e-81
- **Verdict:** ✅ **Reject H_0** — Non-residential properties average $2.89M versus $917K for residential, confirming these as fundamentally different markets.

Test 3 — Chi-Square: Building Class × Price Bracket

- **χ^2 statistic:** 8,054.63
- **p-value:** ≈ 0
- **Verdict:** ✅ **Reject H_0** — Building class and price bracket are strongly associated, meaning property classification directly predicts the price tier it falls into.

Regression Analysis:-

Simple Model (Sale Price ~ Gross Sq. Ft.)

- **Slope:** +$248.46 per additional sq ft
- **R^2:** 0.3526 — size alone explains approximately 35% of price variance
- **RMSE:** $1,133,347

Multiple Model (+ Building Age, Residential Units, Commercial Units)

- **R^2:** 0.3726 | **Adjusted R^2:** 0.3725
- **RMSE:** $1,115,723
- **Improvement over simple model:** +2% R^2

Feature	Standardized Impact
gross_square_feet	+$861,602 (dominant)
building_age	+$184,224
residential_units	−$83,599
commercial_units	−$7,235

The minimal improvement (+2%) when adding three extra features indicates that the missing predictive power resides in categorical variables (like borough and building class) and non-linear relationships, not in more numeric features.**Bottom Line (5 Takeaways)**

1. **All three hypothesis tests rejected the null** — borough, property type, and building class all significantly affect pricing.
2. **Size explains 35% of price**, which appears to be a ceiling for linear models without location encoding.
3. **Non-residential properties are on average three times more expensive** and must be modeled separately.
4. **Building class predicts price tier** (χ^2 = 8,054) and should be a core feature in any predictive model.
5. **Next Step:** A segmented model (per-borough) or tree-based model (Random Forest/XGBoost) is required to overcome the R^2 = 0.37 barrier.

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7. Tableau Dashboard

The final output of the project is an interactive Tableau dashboard, designed to serve as a user-friendly tool for decision-makers.

Dashboard Structure:

The dashboard, titled "NYC Real Estate Market Navigator," is organized into three main tabs:

1. **Market Overview:** High-level KPIs and trends for all of NYC.
2. **Borough Deep Dive:** Detailed comparative analysis across the five boroughs.
3. **Property Type Analysis:** Focus on specific building classes and their valuation metrics.

KPIs, Charts, and Filters Used:

Element	Description	Purpose
KPI Cards	Displays Median Sale Price, Total Sales Volume, and PPSF.	Provides immediate, non-technical understanding of market status.
Map View	Choropleth map of NYC neighborhoods, colored by PPSF.	Geospatial visualization to quickly identify high- and low-value regions.
Trend Line Chart	Monthly Sales Volume and Median Price over the past 12 months.	Highlights seasonality and short-term market momentum.
Filters	`BOROUGH`, `NEIGHBORHOOD`, `PROPERTY TYPE`, and `SALE YEAR`.	Enables end-users to customize the data view for their specific needs.
[A wireframe diagram of the Tableau dashboard layout]		

Support for Decision-Making:

The dashboard supports strategic decision-making by:

- **Identifying Investment Hotspots:** An investor can filter by `PROPERTY TYPE` (e.g., Multi-Family) and use the map view to pinpoint neighborhoods with increasing PPSF and high Sales Volume.
- **Valuation Benchmarking:** A developer can select a specific `BOROUGH` and `PROPERTY TYPE` to obtain the current Median PPSF, providing a quick benchmark for project cost/revenue modeling.
- **Optimizing Sale Timing:** The Trend Line Chart clearly shows peak selling seasons, guiding sellers to list their properties optimally.

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8. Key Insights (8–12 points)

These insights are formulated as actionable statements for a business audience.

1. **Market Distortion by Luxury:** The "average" NYC property is misleading; the mean sale price ($1.07M) is 67% higher than the median ($636K), showing a market heavily skewed by luxury transactions in the upper price tail.

2. **Location Triples Cost:** Median price per square foot in Manhattan is $695, which is 3 times the price in the Bronx ($226)—demonstrating that location alone drastically impacts the cost of space.

3. **The Bulk of the Market:** The $250K–$1M band accounts for 62% of all transactions, representing the standard residential market, while the luxury segment (over $2.5M) makes up only 8% of transactions but heavily skews overall market averages.

4. **Size vs. Location:** While property size scales price predictably, location sets the price ceiling. A "Very Large" property's median price is $4.98M, yet a "Large" property in the Bronx still sells for less than a "Medium" property in Manhattan.

5. **Seasonal Volume, Stable Price:** Summer drives transaction volume (June sees 37% more transactions than February: 5,841 vs 4,276), but median prices remain stable year-round ($600K–$714K), indicating sellers choose to hold through slow months rather than discount.

6. **Age Doesn't Affect Value:** 42.5% of properties are over 80 years old, but building age has a near-zero correlation with price ($r = +0.017$). Pre-war buildings in prime locations retain high premiums.

7. **Density as a Value Driver:** Manhattan's median Floor Area Ratio (FAR) of 2.86 is 5.6 times that of Staten Island (0.51), and FAR correlates positively with sale price ($r = +0.51$)—indicating that denser buildings are worth more.

8. **Divergent Non-Residential Market:** Non-residential properties sell at a lower median price ($465K) compared to residential ($694K), but commercial outliers push the non-residential mean far higher, confirming these are two fundamentally different markets in the same dataset.

9. **Closing Day Pattern:** 24% of all sales close on Thursdays, making it the most common closing day, with Monday being the slowest. This reflects business-hour legal and financial closing patterns.

10. **Size is Necessary but Insufficient:** Square footage is the strongest single predictor of price ($r = +0.59$), but it still leaves 41% of variance unexplained. Accurate predictive models must also incorporate location, building class, and density features.

9. Business Recommendations (3–5)

1. **Adopt Borough-Level Pricing Models Instead of City-Wide Averages**
    a. **Data Evidence:** ANOVA confirmed statistically significant price differences across boroughs ($F = 2{,}607$, $p \approx 0$). Manhattan's median price per sqft ($695) is 3 times the Bronx ($226).
    b. **Recommendation:** Real estate firms should abandon single city-wide valuation models. Implement borough-specific pricing engines that use localized price-per-sqft benchmarks, as one model cannot effectively serve both Manhattan and the Bronx.
2. **Time Listings to Capture the Summer Volume Surge**
    a. **Data Evidence:** June records 37% more transactions than February (5,841 vs 4,276), yet median sale prices remain stable year-round ($600K–$714K). Sellers choose to hold through slow months rather than discount.
    b. **Recommendation:** Sellers and agents should list properties in April–May to maximize exposure to peak buyer demand in June–July. Buyers seeking negotiation leverage should target January–February for reduced competition, not for lower prices.
3. **Invest in Pre-War Building Renovation Over New Mid-Tier Construction**
    a. **Data Evidence:** Building age has near-zero linear correlation with price ($r = +0.017$). Pre-war properties (80–120 years old) in prime locations command luxury-tier pricing comparable to new developments.
    b. **Recommendation:** Developers should evaluate the acquisition and renovation of pre-war buildings in Manhattan and Brooklyn as an alternative to ground-up construction. Renovation can achieve comparable per-sqft sale prices at a potentially lower cost than new construction.
4. **Separate Commercial and Residential Portfolios in All Analysis**
    a. **Data Evidence:** The T-Test confirmed a statistically significant price gap between residential (mean $917K) and non-residential (mean $2.89M) properties ($p = 1.35e-81$). Chi-Square testing proved building class directly determines the price bracket ($\chi^2 = 8{,}054$).
    b. **Recommendation:** Any valuation model, report, or investment analysis that mixes residential and commercial properties will yield misleading results. Organizations should maintain separate analytical pipelines and predictive models for each segment to ensure accurate forecasting.
5. **Prioritize Density (FAR) as a Value Indicator for Land Acquisition**
    a. **Data Evidence:** Floor Area Ratio (FAR) shows a moderate positive correlation with sale price ($r = +0.51$), which is stronger than building

age, unit count, or land area. Manhattan's median FAR (2.86) is 5.6 times Staten Island's (0.51).

b. **Recommendation:** Land acquisition teams should prioritize parcels with high allowable FAR or rezoning potential. The data demonstrates that denser buildings command significantly higher total sale prices, representing strong value-add opportunities.

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10. Impact & Value

The insights generated by this analysis offer tangible, quantified value to key stakeholders:

- **Real Estate Investors:** The seasonal analysis (37% volume surge in June, stable median prices) provides clear guidance on optimal listing timing to maximize sale probability. Borough-level PPSF benchmarks ($695 Manhattan vs. $226 Bronx) enable data-driven location selection.
- **Property Developers:** The FAR correlation ($r = +0.51$) and building age analysis ($r = +0.017$) inform land acquisition strategy — high-density potential in underbuilt parcels represents the strongest value-add opportunity, while pre-war renovation offers a cost-effective alternative to new construction.
- **City Planners/Tax Authorities:** The ANOVA-validated borough price disparities ($F = 2{,}607$) and Chi-Square-confirmed building class associations ($\chi^2 = 8{,}054$) provide statistically robust benchmarks for ensuring fairness in property tax assessments and zoning policy.
- **Financial Institutions:** The regression model's RMSE of $1.12M and R^2 ceiling of 0.37 quantify the current limits of linear valuation — institutions should adopt segmented, non-linear models for collateral assessment to reduce mortgage risk.

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11. Limitations

Despite the rigorous analysis, certain limitations must be acknowledged:

- **Exclusion of Non-Market Sales:** The removal of $0 sales (non-arm's length transactions) during cleaning was necessary but excludes a significant portion of original records, potentially biasing the dataset toward market-rate transactions.

- **Lack of Internal Condition Data:** The dataset provides no information on renovation status, interior condition, or amenity level — these are significant non-quantifiable drivers of price that contribute to the 63% unexplained variance in our regression models.
- **Linear Model Ceiling:** The best R^2 achieved was 0.37 with multiple linear regression. This confirms that the true price drivers include non-linear relationships and categorical interactions (borough × building class) that require tree-based or ensemble methods to capture.
- **Single Time Snapshot:** The analysis is based on a rolling 12-month period, which limits the ability to draw conclusions about long-term market cycles, interest rate impacts, or sustained economic trends.

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12. Future Scope

The project can be extended through the following avenues:

1. **Time-Series Forecasting:** Incorporating five to ten years of historical data to build time-series models (e.g., ARIMA) to forecast future Median Sale Prices and Sales Volumes.
2. **External Data Integration:** Integrating macroeconomic variables (e.g., mortgage interest rates, employment rates) to build a more comprehensive predictive model.
3. **Advanced Geospatial Modeling:** Utilizing spatial autocorrelation techniques to analyze the neighborhood effect on pricing (e.g., the distance to subway lines or parks).

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13. Conclusion

This project set out to address a fundamental challenge in the NYC real estate market: the lack of a clear, data-driven framework for understanding what drives property prices across five boroughs with vastly different dynamics. Using 84,548 property transactions from the NYC Department of Finance, we built a structured analytical pipeline — cleaning and engineering 10 new features in Python, validating patterns through three hypothesis tests (ANOVA, T-Test, Chi-Square), and quantifying predictive relationships via regression modelling. The single most important finding is

that **NYC is not one market — it is at least five.** Manhattan's price per square foot ($695) is 3× the Bronx's ($226), residential and commercial properties operate as fundamentally separate markets (T-Test p = 1.35e-81), and building class directly predicts price tier (χ^2 = 8,054). Yet the market also shows surprising stability: median prices remain flat year-round despite a 37% summer transaction surge, and building age has near-zero impact on value. Based on these findings, the highest-impact action for any stakeholder — whether investor, developer, lender, or policymaker — is to **abandon city-wide averages and adopt borough-specific, segment-specific models** for all pricing, valuation, and policy decisions.

This project delivered a cleaned, analysis-ready dataset (57,601 rows, 31 features), 10 statistically validated insights, 5 actionable business recommendations, a baseline regression model with quantified accuracy (R^2 = 0.37, RMSE = $1.12M), and an interactive Tableau dashboard for ongoing stakeholder exploration.

The R^2 ceiling of 0.37 also establishes a clear direction for future work: the remaining 63% of price variance resides in non-linear interactions, categorical features, and external data (geospatial, economic) that require ensemble methods like Random Forest or XGBoost to capture.

The foundation built here — clean data, validated hypotheses, and a defined KPI framework — makes that next step immediately actionable.

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14. Contribution Matrix

This section must clearly document each team member's contribution across all project phases. Claims must match evidence in GitHub Insights, PR history, and committed files. Any mismatch may result in individual grade adjustments.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Anuj Chhajed	*support*	*support*	*Owner*	*support*	*support*	*support*	*Owner*
Amrit Kumar Mahto	*Owner*	*support*	*support*	*support*	*Owner*	*support*	*support*
Nakul Sharma	*support*	*support*	*support*	*support*	*support*	*Owner*	*support*
Ayush Kumar	*support*	*Owner*	*support*	*support*	*support*	*support*	*support*
Parth Tandalwade	*support*	*support*	*support*	*Owner*	*support*	*support*	*support*